

Swati Singh

(929)420-9626 | ss20478@nyu.edu | [Linkedin](#) | [Github](#)

EDUCATION

New York University

Master of Science, Computer Science

New York, NY

September 2025 – May 2027

Vellore Institute of Technology

Bachelor of Technology, Computer Science and Engineering

India

July 2018 – May 2022

TECHNICAL SKILLS

Languages: Java, Python, C++, SQL, JavaScript, HTML/CSS

Frameworks & AI/ML: Spring Boot, FastAPI, React, Next.js, Machine Learning, NLP, LLM APIs, TensorFlow, Scikit-learn, XGBoost, SHAP, Pandas, NumPy, OpenCV

Tools & Technologies: Docker, Git, PostgreSQL, MySQL, REST APIs, Microservices, CI/CD, Azure DevOps, Google Cloud, Linux, Distributed Systems, GPU Computing

EXPERIENCE

NYU School of Global Public Health

New York, NY

Summer Research Assistant

May 2026 – August 2026

- Working on clinical risk prediction models integrating insurance claims with neighborhood-level social determinants of health to quantify their impact on ER and primary care utilization.
- Analyzing MESA and Jackson Heart Study cohorts using statistical modeling and mediation analysis to study neighborhood effects on cardiovascular disease incidence.

Bajaj Finserv Pvt. Ltd.

Pune, India

Senior Software Engineer

August 2024 – July 2025

- Engineered Java REST APIs and batch jobs using Actico Modeler to support real-time authorization decisions with sub-2ms response times.
- Implemented business rule enhancements for EMI Card authorization workflows, contributing to a 20% increase in approval rates.

Software Engineer

July 2022 - July 2024

- Built a multi-threaded, scalable Java Scheduler for ingesting millions of transaction documents into a DMS system, ensuring data reliability and parallel performance.
- Developed a microservice-based HRMS portal with a maker-checker workflow for secure, validated Excel-based master data management supporting candidate offer-generation APIs.

Software Engineering Intern

Jan 2022 - June 2022

- Modernized legacy Core Java codebases to Spring Boot, improving maintainability and scalability.
- Collaborated with QA Team for Partner Integration Rest APIs reducing Partner Onboarding time from 6 months to 1 month through optimized data handling and testing.

PROJECTS

Foreign Whispers | [FastAPI](#), [Next.js](#), [Docker](#), [Whisper](#), [Chatterbox TTS](#), [GPU Inference](#)

- Built an end-to-end AI video dubbing platform that transforms YouTube videos into dubbed outputs using speech transcription, translation, voice generation, and subtitle alignment.
- Designed containerized FastAPI, Next.js, and GPU inference services for transcription, diarization, TTS generation, subtitle alignment, and video stitching.

CinemaMatch NYC | [React](#), [Python](#), [Node.js](#), [Gemini Vision](#), [NYC Open Data](#), [Google Cloud](#) | [Github](#)

- Built and deployed a Google Cloud-hosted AI platform that analyzes film metadata and recommends hyper-local NYC audience segments using demographic, cultural, and neighborhood datasets.
- Designed audience-matching and go-to-market strategy workflows with geographic heatmaps, engagement scores, and campaign recommendations for independent filmmakers.

Heart Disease Prediction using a Two-Level Hybrid Model | [C++](#), [OpenMP/MPI](#), [MLPack](#) | [Github](#)

- Enhanced a research-based heart disease prediction model by combining Decision Trees with Genetic Algorithms to improve diagnostic performance.
- Integrated parallel computing using OpenMP/MPI to improve training efficiency and computational scalability.

Urban Sound Classification | [Python](#), [CNNs](#), [Librosa](#), [Scikit-learn](#) | [Github](#)

- Built audio preprocessing and deep learning pipelines using MFCCs, spectral features, and CNNs to classify real-world urban sounds from the UrbanSound8K dataset.